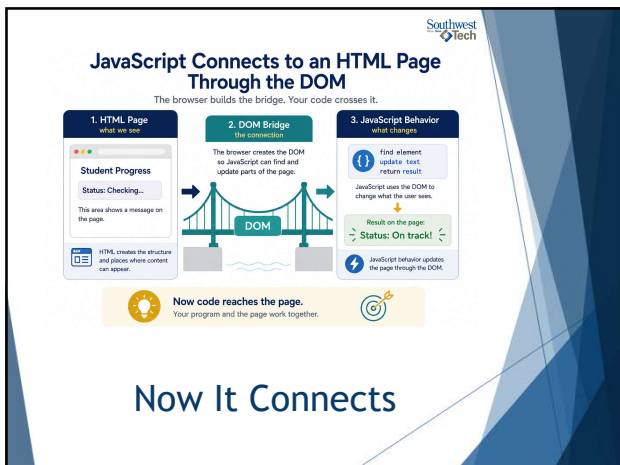




1



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Lab Review: Successful Path

One successful path:

- ▶ variables stored values
- ▶ conditions chose paths
- ▶ functions organized logic
- ▶ console output showed the result
- ▶ no page update was required yet

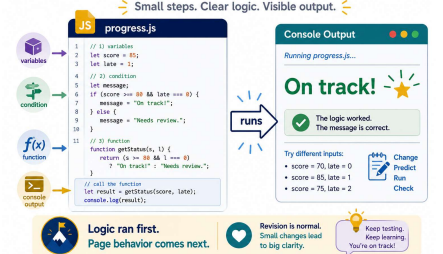
Today the result moves onto the page.



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One Successful Week 4 Console-Only JavaScript Path

Small steps. Clear logic. Visible output.



Logic ran first. Page behavior comes next.

Revision is normal. Small changes lead to big clarity.

Keep testing. Keep learning. You're on track!

Week 4 Console Success Path

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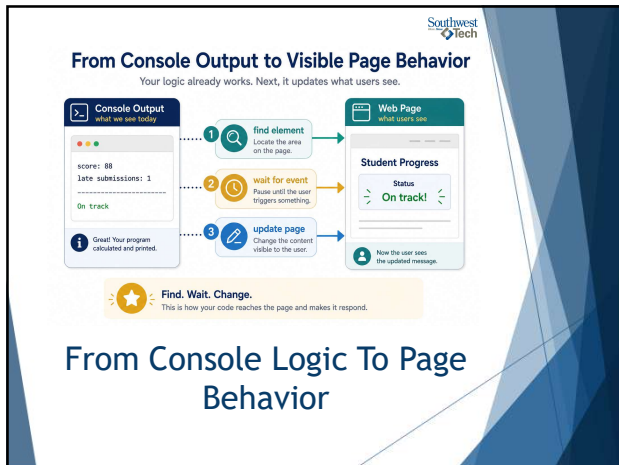
From: Console Logic To: Page Behavior

Console output is useful for learning.
Page behavior is useful for users.

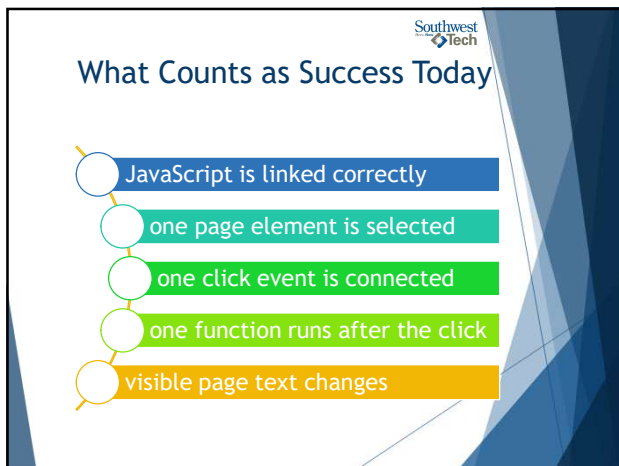
The browser needs three pieces:

- ▶ an element to find
- ▶ an event to wait for
- ▶ code that changes something visible

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Today's Beginner DOM Connection Toolbox

Simple tools. Clear purpose. Connect your code to the page.

These tools help your code find the right element, listen for an action, and update what users see.

DOM Connection Toolbox

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What We Will “Skip” For Now

- ▶ many selector patterns
- ▶ every event type
- ▶ deep form validation
- ▶ jQuery as implementation model
- ▶ saving data
- ▶ complex UI state

Today:
find one element,
respond to one click.

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DOM and Event Topics Parked for Later in Week 5

Important topics. Not today.

P Parked for Later
We'll return to those in Week 5.

- 1. many selectors
More ways to find elements on the page.
- 2. every event type
Lots of events. We start with click.
- 3. full validation
Complete input checks and user feedback.
- 4. jQuery implementation
A library that can do many things.
- 5. saving data
Storing data for later use (e.g., storage, database).
- 6. complex UI state
Multi-step interfaces, many variables, and advanced patterns.

Not today.
First, one click changes one visible result.
Small steps build strong understanding.

Parked For Later

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The DOM Is The Bridge

The DOM is the browser's page model.
It lets JavaScript:

- ▶ find HTML elements
- ▶ read current page values
- ▶ change text or styles
- ▶ respond when users act

The HTML is the source.
The DOM is what JavaScript can work with.

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The DOM: Your Browser's Page Model

The DOM turns your HTML into something JavaScript can use.

HTML source
What you write

```
<!DOCTYPE html>
<html>
  <head>
    <title>My Page</title>
  </head>
  <body>
    <h1 id="title">Welcome!</h1>
    <p id="message">Ready.</p>
    <button id="go">Go</button>
  </body>
</html>
```

DOM page model
What the browser builds

JavaScript can work here
Read, listen, and change

```
// Get items from the DOM
const title =
  document.querySelector('title');
const message =
  document.querySelector('message');
const btn =
  document.querySelector('#go');

// Listen for a click
btn.addEventListener('click', () => {
  // Change the page content
  message.textContent = 'Let's go!';
});
```

The DOM is the bridge JavaScript uses.
HTML becomes a model. JavaScript needs to listen to it,
and updates it — so users see the changes.

DOM Bridge

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Select, Listen, Change

Beginner DOM pattern:

1. select the element
2. listen for an event
3. run a function
4. change something visible

Read DOM code as a connection chain.

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DOM Interaction Chain (Beginner)

JavaScript connects to the page and makes things happen.

1 select element
Find the part of the page you want to work with.
`id="message"`

2 listen for event
Wait for the user to do something (like a click).
`Click Me`

3 run function
Execute the function that handles the event.
`function update() {
 // your logic
}`

4 change visible result
Update the page so the user sees the change.
`Message changed!`

Read DOM code as a connection chain.
Follow the steps from left to right to see how your code connects actions to visible changes on the page.

Select Listen Change Chain

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Selectors Find Existing Elements

`querySelector()` finds an element.

Example:
`const message = document.querySelector("#message");`

Read it as:
"Find the element with id message and remember it."

Selecting does not change the page yet.

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Events Wait For User Action

An event is something that happens.

For today:

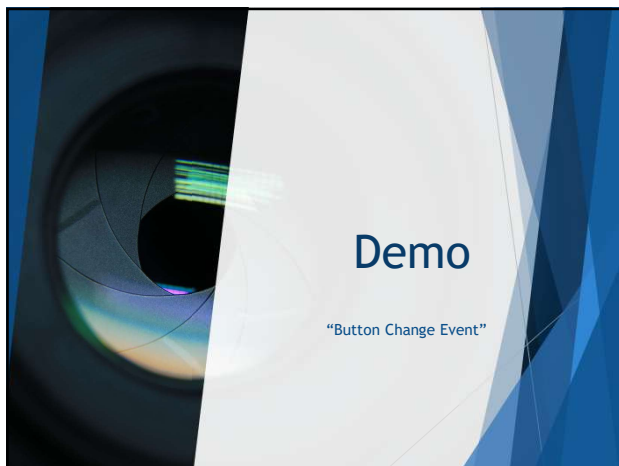
- ▶ user clicks the button
- ▶ browser notices the click
- ▶ JavaScript runs the function
- ▶ the function updates the page

The function waits until the event happens.

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What to Watch For

- ▶ the page before interaction
- ▶ ``#message`` selected
- ▶ ``#updateButton`` selected
- ▶ function changes ``textContent``
- ▶ click event runs the function

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Live Demo: Button Text Change
Click the button. Watch the message change.

Demo:
Button Text Change

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Trace The Connection

Trace the interaction:

HTML:

- ▶ ``id="message"``
- ▶ ``id="updateButton"``

JavaScript:

- ▶ select both elements
- ▶ define the update function
- ▶ listen for a click
- ▶ change ``textContent``

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HTML IDs and JavaScript Selectors
JavaScript finds elements by their id.

Trace HTML To JavaScript Connection

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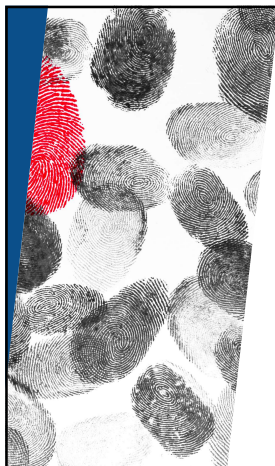
Lab

One visible interaction
Use your existing site.

Your goal:

- ▶ link a separate JavaScript file
- ▶ choose one button, link, or input
- ▶ update text or change a visible element
- ▶ keep the behavior small enough to explain

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Evidence

Preserve evidence:

- ▶ HTML, CSS, and JS files render together
- ▶ JavaScript is in a separate file
- ▶ one user action changes the page
- ▶ the console has no errors
- ▶ you can explain select, event, update

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Closing Success Check

Today:

- ▶ JavaScript found HTML elements
- ▶ a click triggered code
- ▶ a function changed visible text
- ▶ the page became reactive

Next:

- ▶ the page responds to user-provided input.

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